



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 5 – Determinants

Lecture 5.1 – Determinants in terms of permutations

Lecturer: Dr Sean Gardiner – sean.gardiner@unsw.edu.au

Permutations

Recall that a **permutation** on n elements is an ordered arrangement of n distinct objects, or the action that converts one ordering to another. Recall also that the **symmetric group** of order $n!$ is the group S_n of all such actions on n objects. For example, the permutation that converts the tuple $(1, 2, 3)$ to $(2, 1, 3)$ can be written as $\sigma \in S_3$ where $\sigma(1) = 2$, $\sigma(2) = 1$, and $\sigma(3) = 3$.

For a given $\sigma \in S_n$, a convenient way to encode the action of σ is to write it as the list $[\sigma(1), \sigma(2), \dots, \sigma(n)]$. So the permutation that swaps the first two of three ordered elements can be written as $[2, 1, 3]$.

Definition. For a given permutation $\sigma \in S_n$, an **inversion** is any pair i, j between 1 and n for which $i < j$ but $\sigma(i) > \sigma(j)$.

For example, there is 1 inversion in the permutation $[2, 1, 3]$, and 4 inversions in the permutation $[1, 4, 5, 2, 3]$ (for the pairs $(2, 4)$, $(3, 4)$, $(2, 5)$, and $(3, 5)$).

Definition. The **sign** of a permutation σ , denoted $\text{sign}(\sigma)$, is given by $\text{sign}(\sigma) = (-1)^k$ where k is the number of inversions in σ .

For example, $\text{sign}([2, 1, 3]) = -1$ and $\text{sign}([1, 4, 5, 2, 3]) = 1$.

Properties of permutation sign

Lemma. For any permutation $\sigma \in S_n$, we have

$$\text{sign}(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(i) - \sigma(j)}{i - j}.$$

Proof. Since σ is a permutation, for each choice of i and j , either $(i - j)$ or $(j - i)$ appears in the numerator product. In each case that $(j - i)$ appears in the numerator, this corresponds with some inversion in σ . So after cancellation, the right-hand side is -1 multiplied by itself exactly the number of times that there are inversions in σ , which is the sign of σ .

Lemma. For all $\sigma, \tau \in S_n$, we have $\text{sign}(\sigma \circ \tau) = \text{sign}(\sigma) \text{sign}(\tau)$.

Proof. We can write $\text{sign}(\sigma \circ \tau)$ as

$$\begin{aligned} \prod_{1 \leq i < j \leq n} \frac{\sigma(\tau(i)) - \sigma(\tau(j))}{i - j} &= \prod_{1 \leq i < j \leq n} \frac{\sigma(\tau(i)) - \sigma(\tau(j))}{\tau(i) - \tau(j)} \prod_{1 \leq i < j \leq n} \frac{\tau(i) - \tau(j)}{i - j} \\ &= \prod_{1 \leq i < j \leq n} \frac{\sigma(i) - \sigma(j)}{i - j} \prod_{1 \leq i < j \leq n} \frac{\tau(i) - \tau(j)}{i - j} \end{aligned}$$

since τ is a permutation. Note that the right-hand side is $\text{sign}(\sigma) \text{sign}(\tau)$.

Corollary. The function $\text{sign} : S_n \rightarrow \{-1, 1\}$ is a group homomorphism.

Determinants

With the language of permutations, we can define the determinant of a square matrix in the following way.

Definition. The **determinant** of a matrix $A \in M_{n,n}(\mathbb{F})$, denoted $\det(A)$ or $|A|$, is the scalar value

$$\det(A) = \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} = \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$$

where a_{ij} is the (i,j) th entry of A .

We can check that this definition aligns with the one we are more familiar with from first year for some small cases:

Example. For the case $n = 2$, since $S_2 = \{[1, 2], [2, 1]\}$, we have

$$\begin{aligned} \det \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} &= \text{sign}([1, 2]) a_{1,1} a_{2,2} + \text{sign}([2, 1]) a_{1,2} a_{2,1} \\ &= a_{1,1} a_{2,2} - a_{1,2} a_{2,1}. \end{aligned}$$

3×3 determinants

Example. In the $n = 3$ case, let the elements of S_3 be $\sigma_1 = [1, 2, 3]$, $\sigma_2 = [1, 3, 2]$, $\sigma_3 = [2, 1, 3]$, $\sigma_4 = [2, 3, 1]$, $\sigma_5 = [3, 1, 2]$, and $\sigma_6 = [3, 2, 1]$.

Then for $A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix}$ we have by the permutation definition

$$\begin{aligned} \det(A) &= \text{sign}(\sigma_1)a_{1,1}a_{2,2}a_{3,3} + \text{sign}(\sigma_2)a_{1,1}a_{2,3}a_{3,2} + \text{sign}(\sigma_3)a_{1,2}a_{2,1}a_{3,3} \\ &\quad + \text{sign}(\sigma_4)a_{1,2}a_{2,3}a_{3,1} + \text{sign}(\sigma_5)a_{1,3}a_{2,1}a_{3,2} + \text{sign}(\sigma_6)a_{1,3}a_{2,2}a_{3,1} \\ &= a_{1,1}a_{2,2}a_{3,3} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} \\ &\quad + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1}. \end{aligned}$$

This aligns with the more familiar “expand along a row” method, for example:

$$\begin{aligned} \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} &= a_{1,1} \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} - a_{1,2} \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} + a_{1,3} \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix} \\ &= a_{1,1}a_{2,2}a_{3,3} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} \\ &\quad + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1}. \end{aligned}$$

Properties of the determinant

Lemma. For any $A \in M_{n,n}(\mathbb{F})$, we have $\det(A) = \det(A^T)$.

Proof. Each summand in the formula for $\det(A)$ is a product of entries ordered by their row position, but could instead be ordered by their column position like so:

$$\sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} = \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{\sigma^{-1}(1),1} a_{\sigma^{-1}(2),2} \cdots a_{\sigma^{-1}(n),n}.$$

Since we are summing over all permutations (which are bijections), the sum over all $\sigma \in S_n$ is equivalent to taking the sum over all $\sigma^{-1} \in S_n$.

Furthermore, we know $\text{sign}(\sigma^{-1}) = \text{sign}(\sigma)$ since

$$\text{sign}(\sigma) \text{sign}(\sigma^{-1}) = \text{sign}(\sigma \circ \sigma^{-1}) = \text{sign}(\text{id}) = (-1)^0 = 1.$$

So writing $\tau = \sigma^{-1}$, we have

$$\det(A) = \sum_{\tau \in S_n} \text{sign}(\tau) a_{\tau(1),1} a_{\tau(2),2} \cdots a_{\tau(n),n} = \det(A^T).$$

Corollary. For any $A \in M_{n,n}(\mathbb{F})$, we have $\det(A^*) = \overline{\det(A)}$.

(This is by linearity of the complex conjugate.)

Properties of the determinant

Lemma. If the rows (or columns) of a matrix $A \in M_{n,n}(\mathbb{F})$ are reordered by some permutation $\pi \in S_n$ to create the matrix B , then $\det(B) = \text{sign}(\pi) \det(A)$.

Proof. Suppose π permutes the columns of A , so that each entry of B is given by $b_{i,j} = a_{i,\pi(j)}$. (If π permutes the rows instead, consider A^T .) Then writing $\tau = \sigma \circ \pi \in S_n$, we have

$$\begin{aligned} \det(B) &= \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n a_{i,\sigma(\pi(i))} = \sum_{\tau \in S_n} \text{sign}(\tau \circ \pi^{-1}) \prod_{i=1}^n a_{i,\tau(i)} \\ &= \text{sign}(\pi^{-1}) \sum_{\tau \in S_n} \text{sign}(\tau) \prod_{i=1}^n a_{i,\tau(i)}, \end{aligned}$$

where we have used the facts that the sign function is a homomorphism and that $\text{sign}(\pi^{-1})$ is constant over the sum.

Since $\text{sign}(\pi^{-1}) = \text{sign}(\pi)$ (as seen in the previous proof), we have that $\det(B) = \text{sign}(\pi) \det(A)$.

Properties of the determinant

Lemma. If a matrix has two equal rows or columns, then its determinant is zero.

Proof. Suppose the matrix A has two equal rows (or columns) in positions i and j . If σ is the permutation swapping positions i and j only with $i < j$, then we know from the previous lemma that $\det(A) = \text{sign}(\sigma) \det(A)$. So if $\text{sign}(\sigma) = -1$, we must have that $\det(A) = 0$.

To see that $\text{sign}(\sigma) = -1$, note that the only inversions in this permutation must involve i or j . After swapping, j is involved in $j - i$ inversions (j precedes each of $i, i + 1, \dots, j - 1$), and i is also involved in $j - i$ inversions (i succeeds each of $i + 1, i + 2, \dots, j - 1, j$). Amongst these two lists, exactly one pair (i, j) is counted twice. So altogether there are $2(j - i) - 1$ inversions, meaning $\text{sign}(\sigma) = (-1)^{2(j-i)-1} = -1$ since the exponent is always odd.

Properties of the determinant

Lemma. Given a matrix $A \in M_{n,n}(\mathbb{F})$, adding a multiple of a row/column to any other row/column does not change the determinant.

Proof. Suppose that B is the result of adding λ times the j th row to the i th row of A . Then $b_{i,k} = a_{i,k} + \lambda a_{j,k}$ for each $1 \leq k \leq n$. So expanding out the determinant gives (assuming without loss of generality that $i < j$)

$$\begin{aligned}\det(B) &= \sum_{\sigma \in S_n} \text{sign}(\sigma) b_{1,\sigma(1)} \dots b_{i,\sigma(i)} \dots b_{j,\sigma(j)} \dots b_{n,\sigma(n)} \\ &= \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} \dots (a_{i,\sigma(i)} + \lambda a_{j,\sigma(i)}) \dots a_{j,\sigma(j)} \dots a_{n,\sigma(n)} \\ &= \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} \dots a_{i,\sigma(i)} \dots a_{j,\sigma(j)} \dots a_{n,\sigma(n)} \\ &\quad + \lambda \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{1,\sigma(1)} \dots a_{j,\sigma(j)} \dots a_{j,\sigma(j)} \dots a_{n,\sigma(n)} \\ &= \det(A) + 0,\end{aligned}$$

since the second term is the determinant of a matrix with two equal rows.

Multilinearity of the determinant

Notice that we can think of the determinant as a function on each of the columns of an $n \times n$ matrix, that is, as $\det : \underbrace{\mathbb{F}^n \times \cdots \times \mathbb{F}^n}_{n \text{ copies}} \rightarrow \mathbb{F}$.

Lemma. When considered in this way, the determinant is an **alternating multilinear map**.

Proof. To prove multilinearity, we only need to prove linearity in the first argument (column) since linearity in the others follows from the row-swapping properties of the determinant. So consider matrices $A, B \in M_{n,n}(\mathbb{F})$ where B has zero entries in all but the first column. Then

$$\begin{aligned}\det(A + \lambda B) &= \sum_{\sigma \in S_n} \text{sign}(\sigma)(a_{1,\sigma(1)} + \lambda b_{1,\sigma(1)})a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} \\ &= \sum_{\sigma \in S_n} \text{sign}(\sigma)a_{1,\sigma(1)} \cdots a_{n,\sigma(n)} + \lambda \sum_{\sigma \in S_n} \text{sign}(\sigma)b_{1,\sigma(1)}a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} \\ &= \det(A) + \lambda \det(C),\end{aligned}$$

where C is the the matrix A with its first column swapped for the first column of B . The fact that the determinant is an alternating map is clear from the property that permuting the rows/columns has the effect of multiplying the determinant by the sign of that permutation.

Matrix minors

Definition. Given a matrix $A \in M_{n,n}(\mathbb{F})$, the (i, j) minor of A is the matrix $A_{i,j}$ obtained by removing the i th row and the j th column of A .

(Careful: similar notation is sometimes used for the (i, j) th entry of A .)

Lemma. In a matrix $A \in M_{n,n}(\mathbb{F})$, if the entry $a_{i,j}$ is the only nonzero entry in the i th row, then $\det(A) = (-1)^{i+j} a_{i,j} \det(A_{i,j})$.

Proof. Suppose first that $i = j = n$, so that every entry in the last row is 0 except for the final entry $a_{n,n}$. Then in the formula for the determinant, the $a_{n,\sigma(n)}$ term will only be nonzero when $\sigma(n) = n$, so the sum is really only over all permutations of the first $n - 1$ elements, meaning we have

$$\det(A) = a_{n,n} \sum_{\tau \in S_{n-1}} \text{sign}(\tau) a_{1,\tau(1)} a_{2,\tau(2)} \cdots a_{n-1,\tau(n-1)} = a_{n,n} \det(A_{n,n})$$

which matches the expected result since $(-1)^{2n} = 1$.

In the general case for $a_{i,j}$, we can first swap the i th and n th rows, and then the j th and n th columns to achieve the matrix described previously. This has the effect of multiplying the determinant by the sign of both permutations, which are $(-1)^{n-i}$ and $(-1)^{n-j}$ respectively. So in general, we have $\det(A) = (-1)^{2n-i-j} a_{i,j} \det(A_{i,j}) = (-1)^{i+j} a_{i,j} \det(A_{i,j})$.

(Note that this is a special case of the first-year “expand along a row” method.)

Finding determinants

The previous result (which clearly also holds when “row” is replaced by “column”) can be used to help find determinants efficiently. Given any matrix, we can add multiples of rows/columns to other rows/columns until one of the rows/columns contains at most one nonzero entry. Then the formula from the previous lemma reduces the problem to finding the determinant of a smaller matrix, and this process can be repeated until the smaller matrix is 1×1 (or just 2×2 , since the formula for the determinant in that case is straightforward).

Exercise. Find the determinant of $\begin{pmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{pmatrix}$.

Solution. We have

$$\begin{vmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & 1 & -2 & 2 \\ 5 & 2 & -2 & 1 \end{vmatrix} = 2 \begin{vmatrix} 2 & 1 & 2 \\ 1 & -2 & 2 \\ 2 & -2 & 1 \end{vmatrix} = 2 \begin{vmatrix} 0 & 5 & -2 \\ 1 & -2 & 2 \\ 0 & 2 & -3 \end{vmatrix} \\ = 2(-1) \begin{vmatrix} 5 & -2 \\ 2 & -3 \end{vmatrix} = -2(-15 + 4) = 22.$$

Finding determinants from diagonals

Lemma. The determinant of an upper (or lower) triangular matrix is given by the product of its diagonal entries.

Proof. Suppose $A \in M_{n,n}(\mathbb{F})$ is upper triangular, so that whenever $i > j$ the (i, j) th entry $a_{i,j} = 0$. Then the only nonzero summands in the formula for $\det(A)$ are those for which $i \leq \sigma(i)$. In particular, we must have $\sigma(n) = n$, so $\sigma(n-1) = n-1$, and so on, implying the only such σ is the identity permutation. Since the identity permutation has sign $(-1)^0 = 1$, it follows that $\det(A) = a_{1,1}a_{2,2} \dots a_{n,n}$, the product of the diagonals.

The lower triangular case works similarly (or just consider the transpose).

So another efficient method to find the determinant is to add multiples of rows/columns to other rows/columns until the matrix is upper or lower triangular, and then just take the product of the diagonals.

Another useful technique here is to note that swapping any two rows or columns has the effect of changing the sign of the determinant.

Example

Exercise. By converting to an upper or lower triangular matrix, find the

determinant of the matrix $\begin{pmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{pmatrix}$.

Solution. We have

$$\begin{aligned} \begin{vmatrix} 2 & 0 & 2 & 0 \\ 3 & 2 & 4 & 2 \\ 4 & 1 & 2 & 2 \\ 5 & 2 & 3 & 1 \end{vmatrix} &= \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 2 & 1 & 2 \\ 4 & 1 & -2 & 2 \\ 5 & 2 & -2 & 1 \end{vmatrix} = - \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 2 & 2 \\ 4 & -2 & 1 & 2 \\ 5 & -2 & 2 & 1 \end{vmatrix} \\ &= - \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & -2 & 5 & 6 \\ 5 & -2 & 6 & 5 \end{vmatrix} = - \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & -2 & -1 & 1 \\ 5 & -2 & 6 & 5 \end{vmatrix} \\ &= - \begin{vmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & -2 & -1 & 0 \\ 5 & -2 & 6 & 11 \end{vmatrix} \\ &= (-1) \times 2 \times 1 \times (-1) \times 11 = 22. \end{aligned}$$